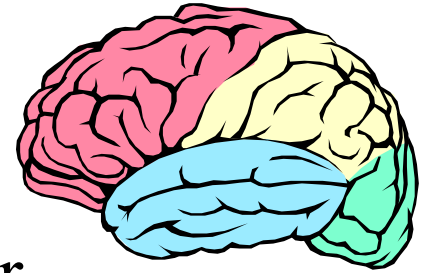


Class 01D: Introduction to ANN

Artificial Neural Networks and Perceptron

Our Amazing Brain !



- The 3 pounds of grey matter houses our central nervous system.
- A neuron is the basic component.
- The human brain incorporates nearly 100 billion (10^{11}) neurons.
- Each neuron has an average of 10,000 connections, **synapses**. (i.e. 10^{15} synapses)
- **Our biological neural network.**

Amazing Brain !

Our brain ? ...3 pounds of grey matter

- The brain consists of a densely interconnected set of nerve cells, or basic information-processing units, called **neurons**.
- Where is the manual ?
- Our brain is performing its functions **much faster** than the fastest computers in existence today.
- A brain's **memory capacity** is comparable to the largest library on earth.

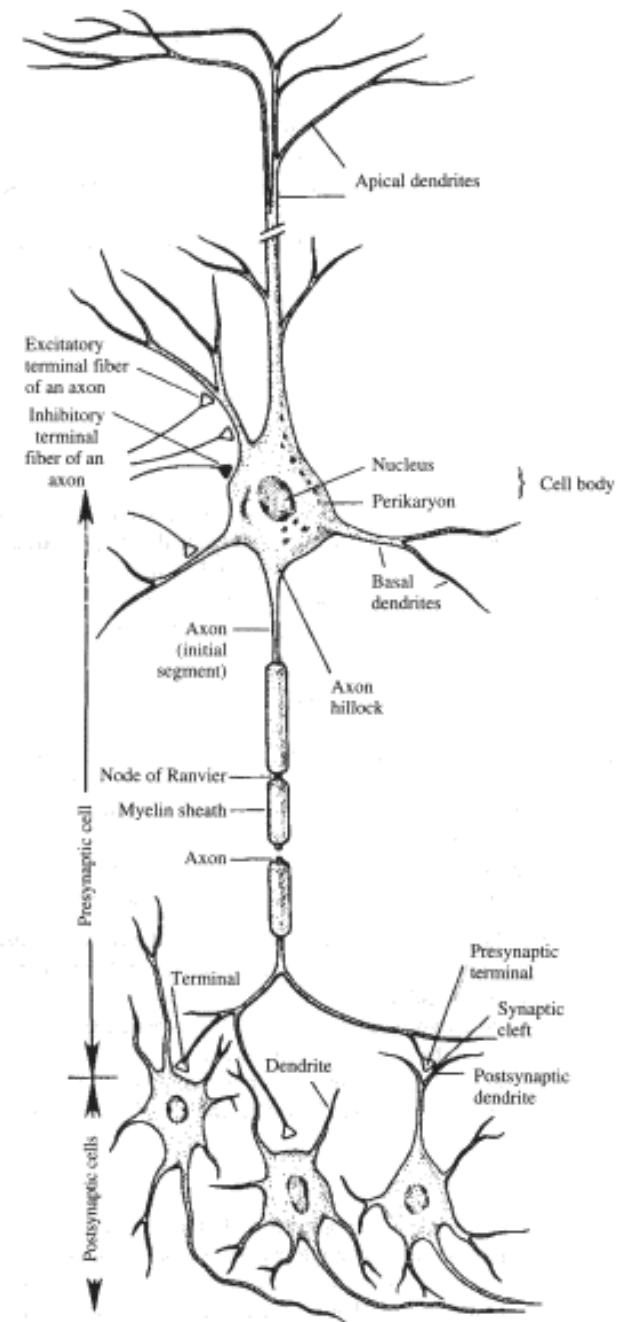


FIGURE 3.1
Expanded view of single neuron morphology [Kan91].

Our Amazing Brain !

- Brain only makes **new cells** during fetal development and for a few months after birth.
- In the first few years of life, more than one million new connections are formed every second!
- Afterwards ... the new information is encoded by **strengthening or making connections** or breaking connections among neurons.
- With pruning, the neuron structures are replaced by more complex and efficient structures.
- **Connections are also related to forming memories.**

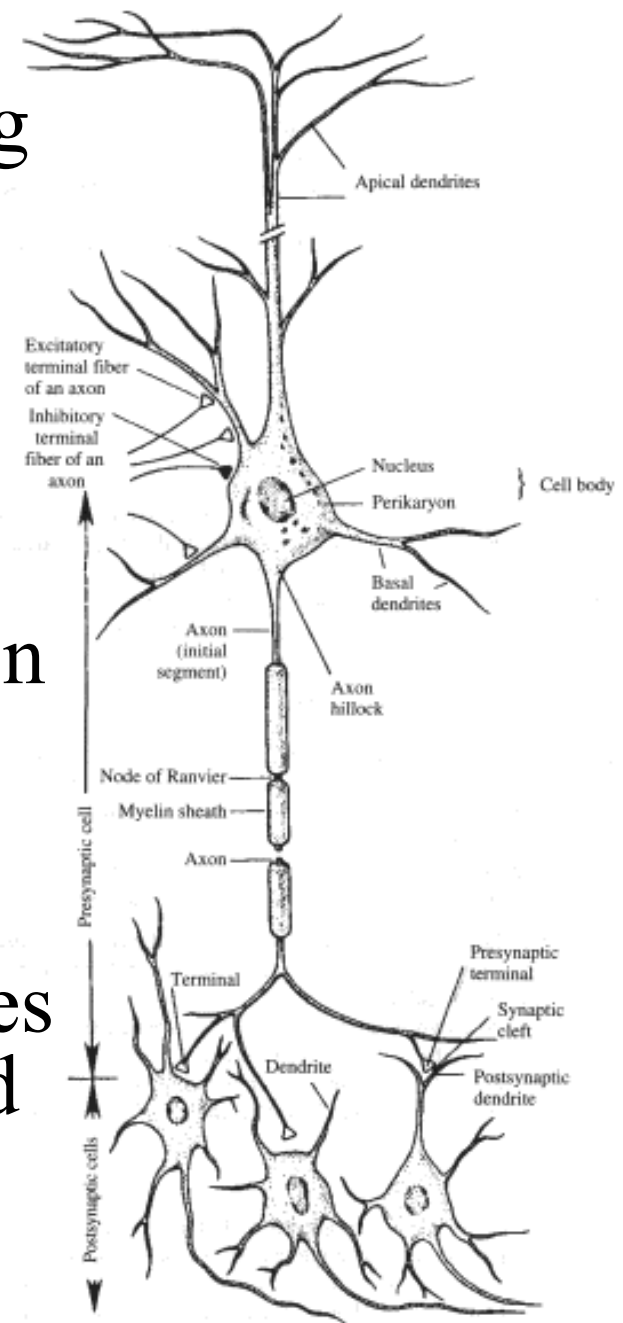
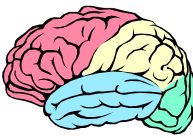
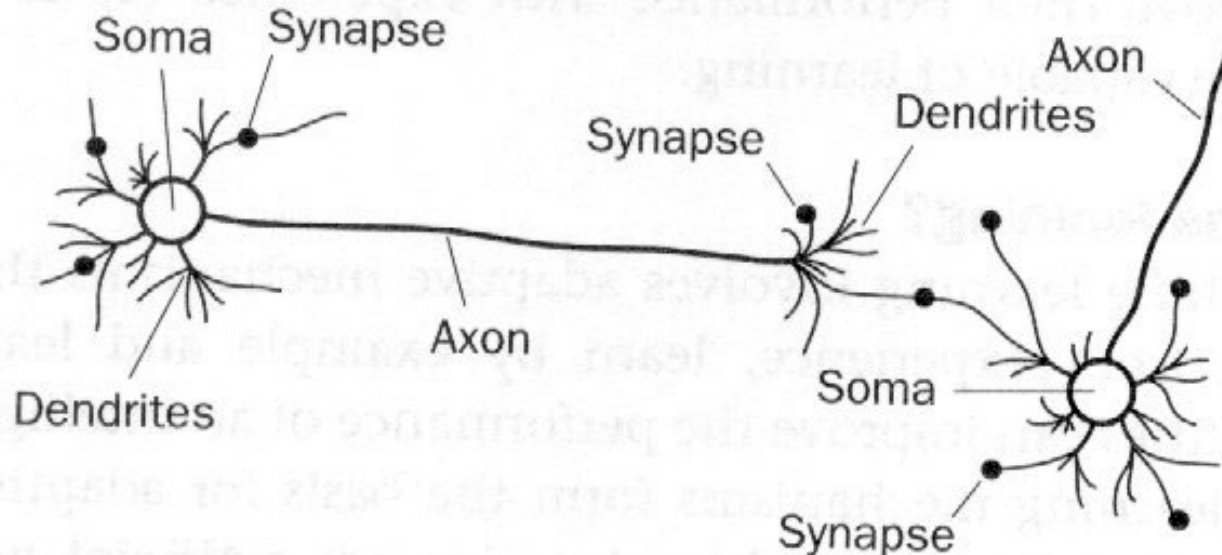


FIGURE 3.1
Expanded view of single neuron morphology [Kan91].

Structure of a neuron



- **Neurons** are the information processing cell of the brain.
- Each neuron has a very simple structure. It consists of a cell body called **soma**, a number of fibres called **dendrites**, and a single long fibre called the **axon**.
- While dendrites branch into a network around the soma, the axon stretches out to the dendrites and somas of other neurons.



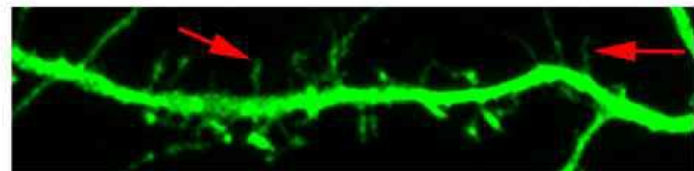
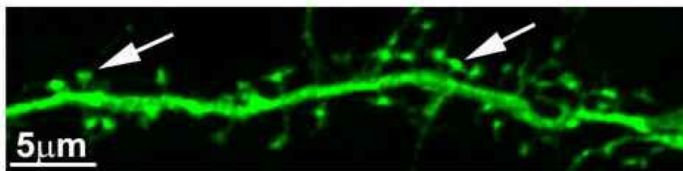
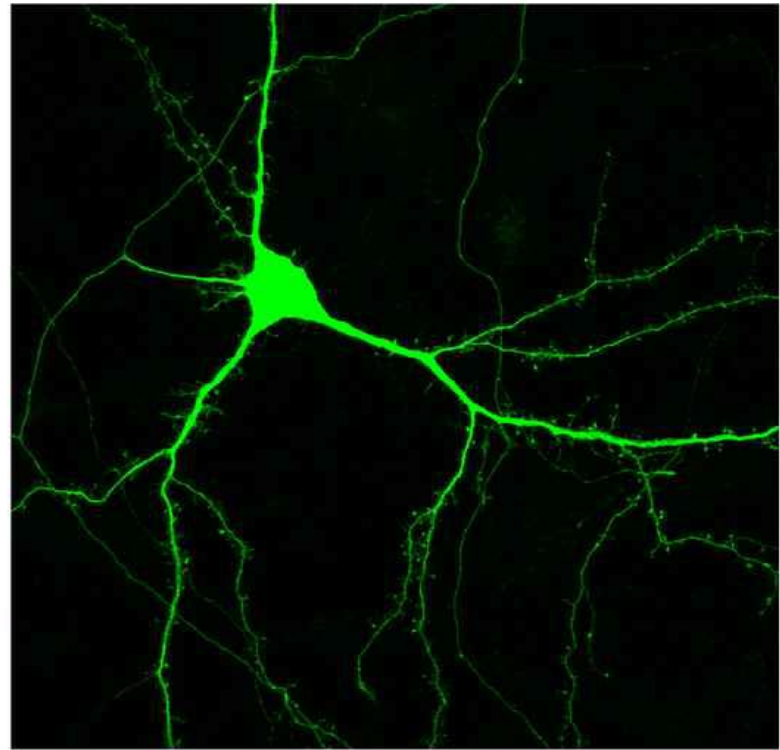
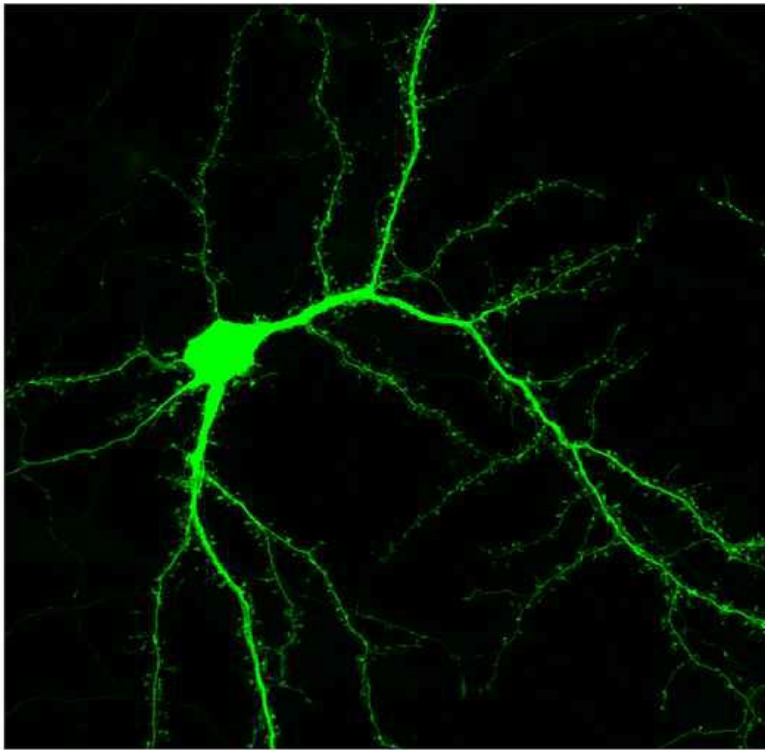
The root of memory

26 August 2016

Dr Lai Kwok-On is intrigued by neuroscience.

Dr. Lai is at the School of Biomedical Sciences (Faculty of Medicine) at HKU. His research aims to understand the signal transduction mechanisms inherent in the development and functioning of neuronal synapses in order to obtain a better understanding of the cellular and molecular basis of long term memory and brain disorders such as autism.

“The brain is so much more complicated than other organs,” he says. The human brain is remarkably flexible. This ‘plastic’ nature of the brain is attributed to the **ability of neurons to change their synaptic connectivity** by experience, therefore contributing to our **ability to learn and form memories.**



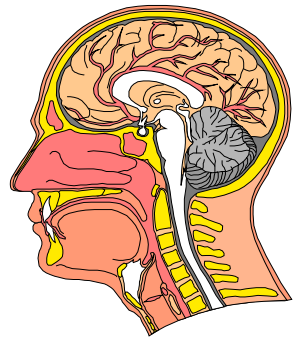
Two rat neurons expressing the green fluorescence protein GFP, illustrating the two types of spines, namely mushroom spines (white arrows) and thin spines (red arrows)

Biological neurons



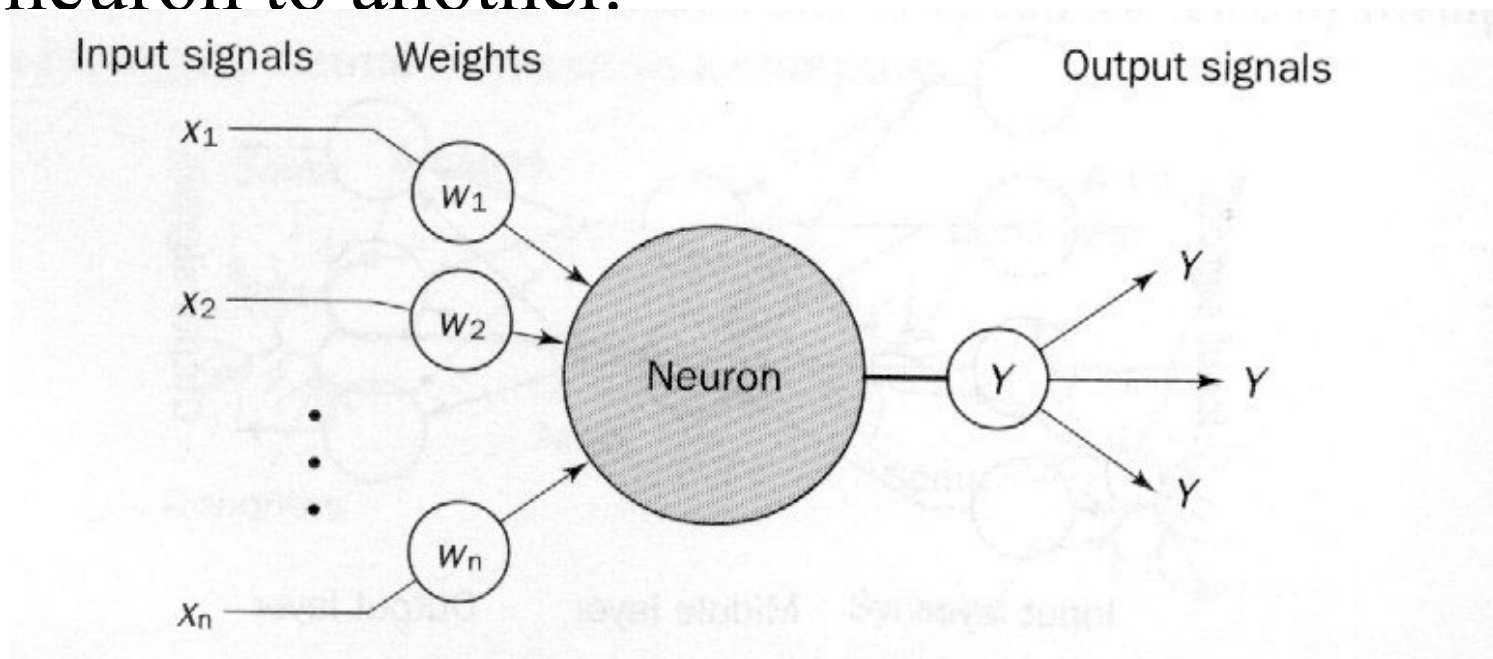
- Each neuron receives its input **electrical signals** along the dendrites.
- From the soma, the axon carries an **electrical signal** away from the soma to another neuron.
- Axons usually terminate on the dendrites of other cells or on muscles.
- There is a difference in electrical potential across the cell membrane which is called **resting potential (around -65mV)**.
- This is due to the difference in **ion concentrations** inside and outside the cell (neuron) which is maintained by the cell membrane. (The ions are sodium, potassium, chloride.)
- A neuron integrates thousands of temporal signals through its dendrites. This **integration** results in a disturbance in the resting potential of the neuron. When the cell potential exceeds a threshold of -40mV, the neuron **fires a pulse** (an action potential) along its axon.

- Hence, signals are propagated from one neuron to another by complex **electro-chemical reactions**. Chemical substances released from the synapses cause a change in the electrical potential of the cell body.
- **The strength of connection between two neurons can change.**
- **Neurons can form new connections with other neurons. Some connections with other neurons may get disconnected.**
- The brain is therefore a highly complex, nonlinear and **parallel information-processing system**.
- Information is stored and processed in a parallel network simultaneously throughout the whole network, rather than at specific locations.
- **Our brain is a biological neural network.** We want to emulate this learning capability using **artificial neural network (ANN)** in a computer.
- A neural network can be defined as a model of reasoning based on the human brain. It is a very popular approach to **machine learning**.

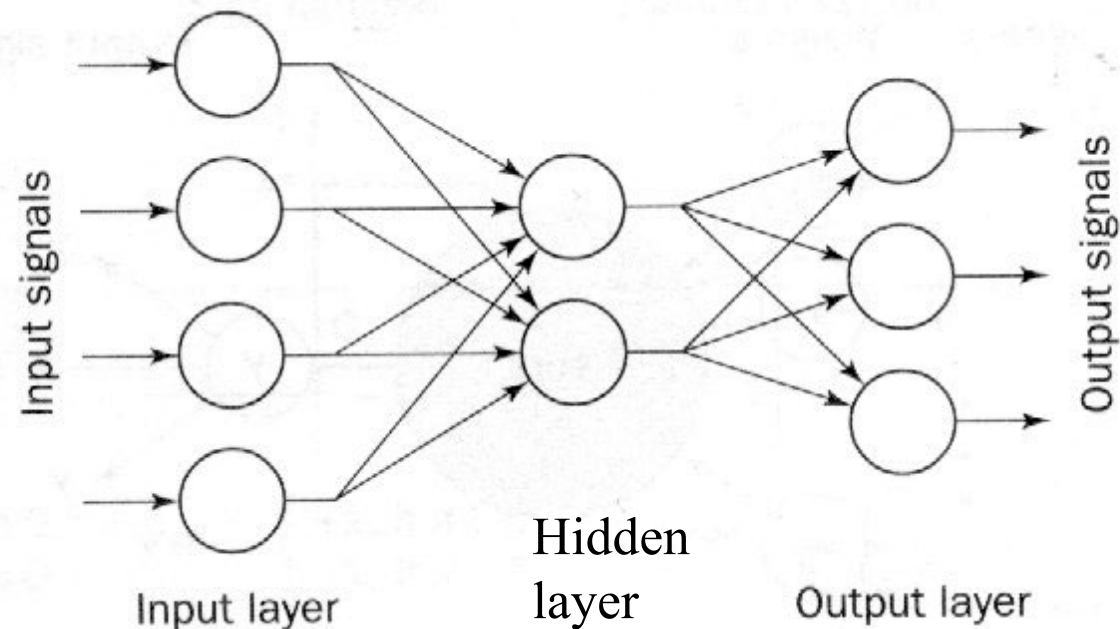


How does **ANN** model the brain ?

- An ANN consists of a number of very simple and highly interconnected processors, also called neurons. The neurons are connected by weighted links passing signals from one neuron to another.



- A neuron receives several signals from its input links, computes to a new activation level and sends it as an output signal through the output links.
- The input signal can be raw data or outputs of other neurons.
- The output signal can be either a final solution to the problem or an input to other neurons. An ANN is shown.



How does an ANN ‘learn’ ?

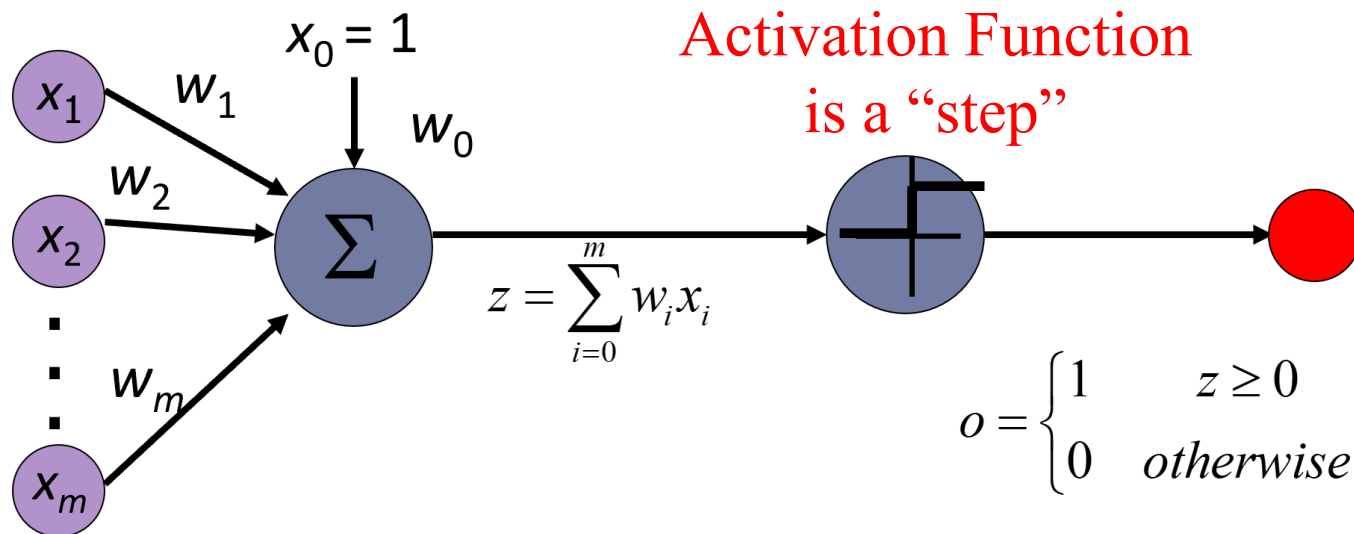
- The neurons are connected by links, and **each link has a numerical weight** associated with it. **Weights** are the basic means of long-term memory in ANNs.
- They express the strength (i.e. the importance) of each neuron input. A neural network ‘learns’ through repeated **adjustments** of these weights.

How to build an ANN ?

- We must first decide how many neurons are to be used and how the neurons are to be connected to form a network. The **number of hidden units and number of hidden layers** would affect the number of parameters of ANN.
- Then, we decide which learning algorithm to use. The **learning rate** can be set.
- Finally, we train the neural network by **initializing** the weights of the network and **update** the weights from a set of training examples.
- The frequency of weight update (**batch size**) can be decided as well.

Perceptrons: the First Neural Network Model

- In 1943, Warren McCulloch and Walter Pitts proposed a very simple idea that is still the basis for most ANN. It is based on a simplified model of a biological neuron.
- **The neuron computes the weighted sum of the input signals and compares the result with a threshold value b .**



The input to the perceptron is

$$z = \sum_{i=1}^m w_i x_i + b = \sum_{i=0}^m w_i x_i$$

The threshold is model by the value “b”. (the old model is “-b”)

The “b” is a parameter, like weights w , that needs to be learned.

The inputs form a vector $\mathbf{x}$ and the vector form of the net input is:

$$z = \mathbf{w}^T \mathbf{x} + b$$

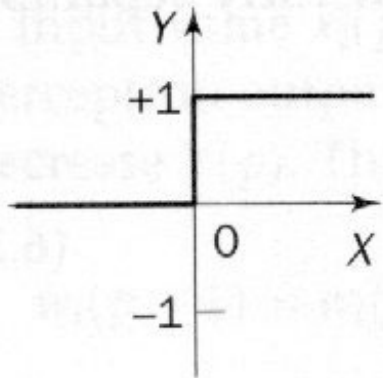
The perceptron fires if and only if the net input is positive

$$z = \mathbf{w}^T \mathbf{x} + b \geq 0$$

Then, the perceptron applies an activation function to the net input and gives an output.

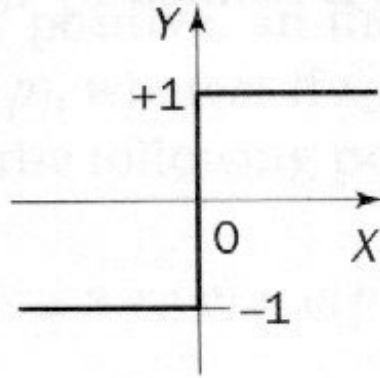
- The type of activation function (in previous slide) is called a **step** function.
- Four common activation functions are:

Step function



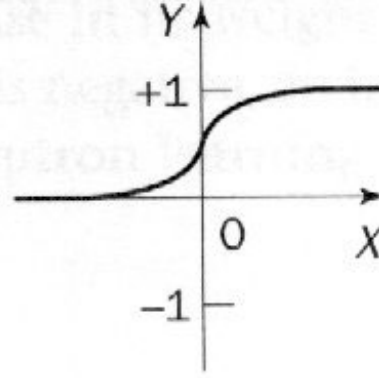
$$Y^{step} = \begin{cases} 1, & \text{if } X \geq 0 \\ 0, & \text{if } X < 0 \end{cases}$$

Sign function



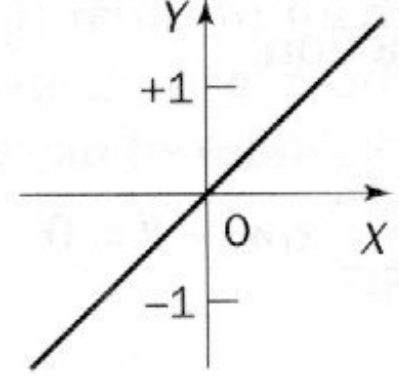
$$Y^{sign} = \begin{cases} +1, & \text{if } X \geq 0 \\ -1, & \text{if } X < 0 \end{cases}$$

Sigmoid function

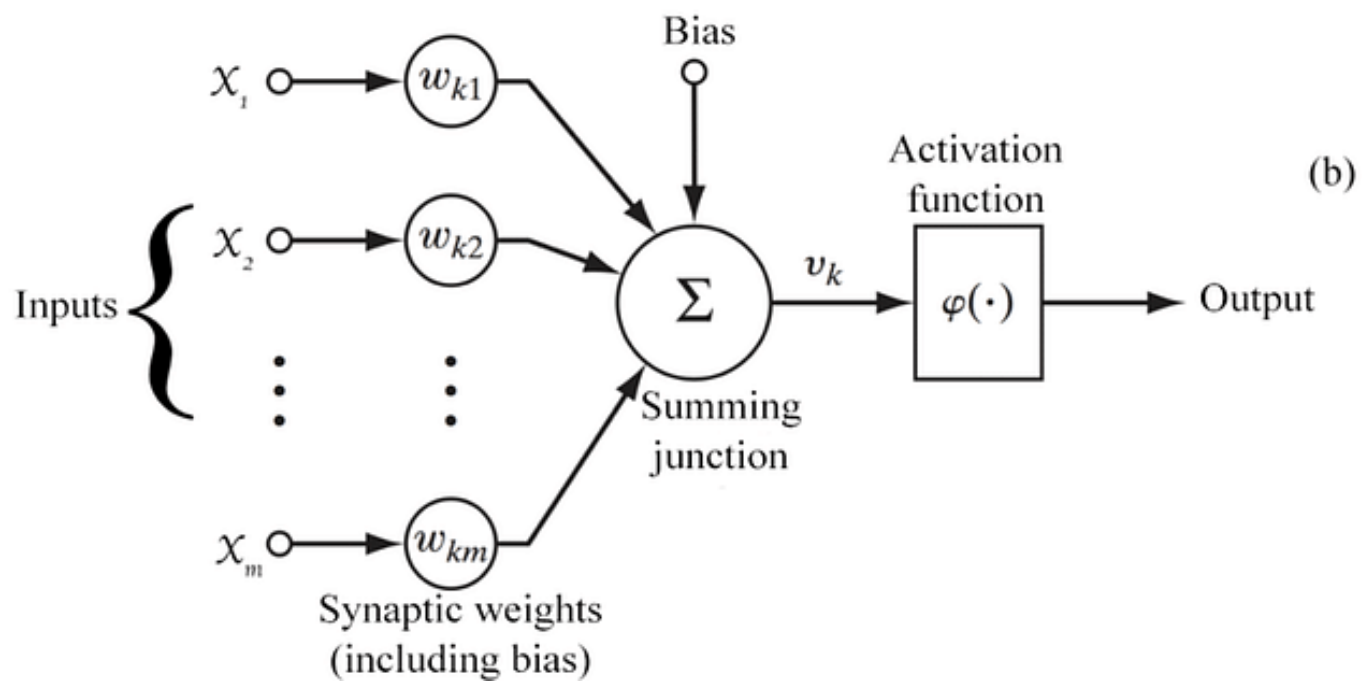
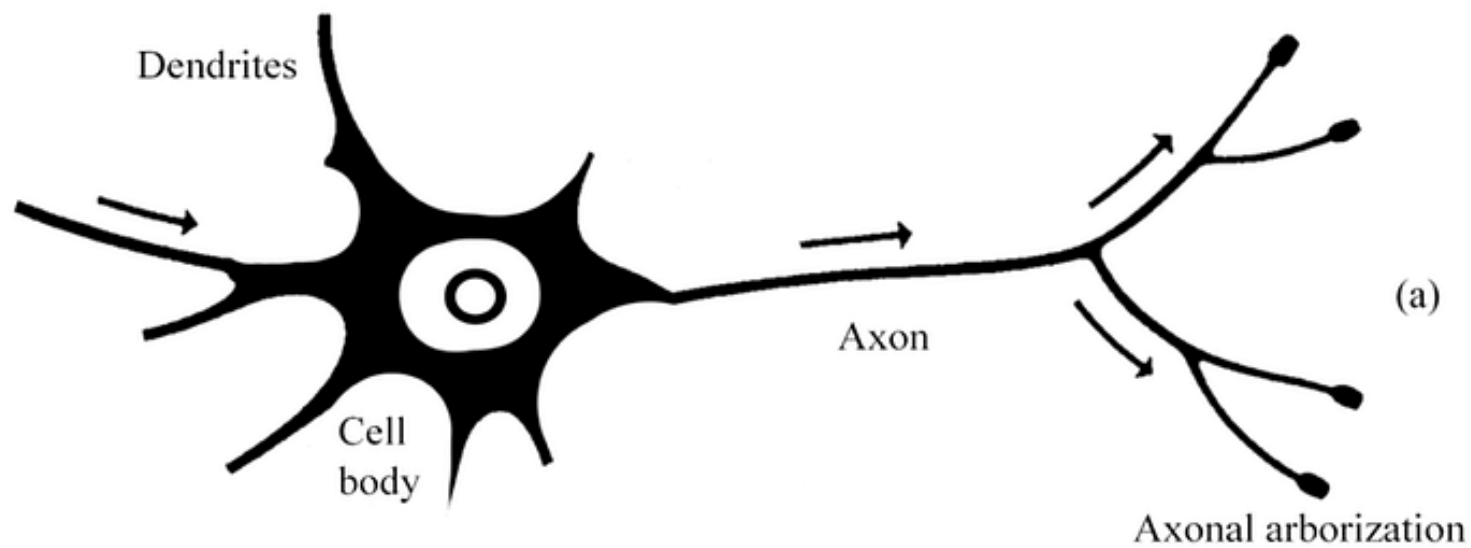


$$Y^{sigmoid} = \frac{1}{1 + e^{-X}}$$

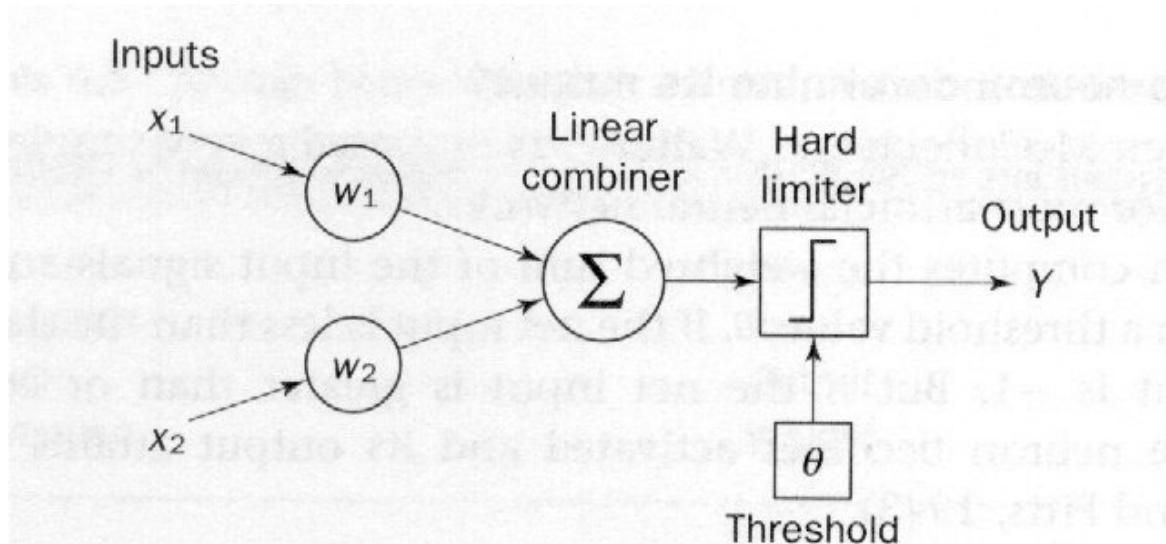
Linear function



$$Y^{linear} = X$$



- The **step** and **sign** activation functions are also called **hard limit** functions. They are often used in decision-making neurons for classification and pattern recognition tasks.
- The **sigmoid** function transforms the input in a range from plus and minus infinity, into a reasonable value in the range between 0 and 1.
- In 1958, Frank Rosenblatt introduced a training algorithm that provided a procedure for training a simple ANN : a perceptron.
- The perceptron is the simplest form of a neural network with **adjustable** synaptic weights and a **hard limiter**. A single-layer two-input perceptron is shown below:



The Perceptron

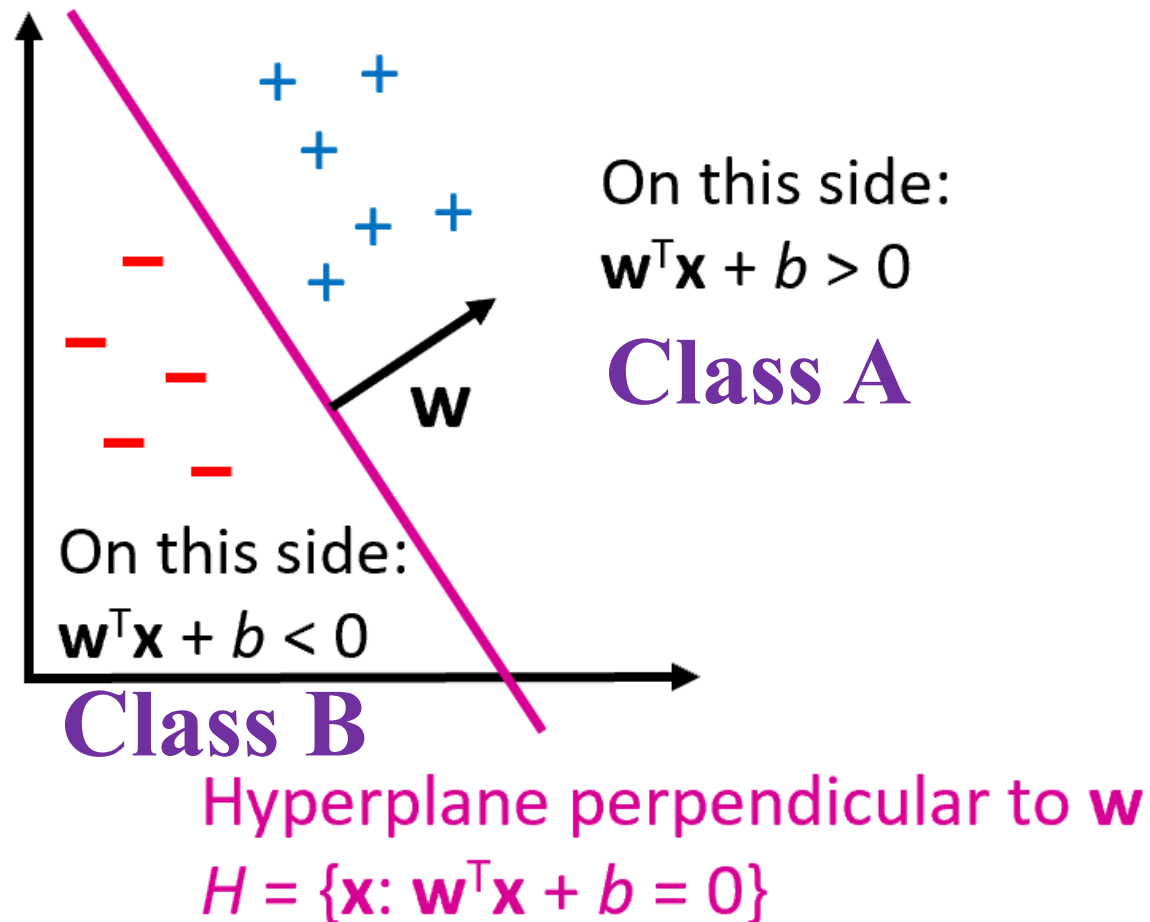
- The operation of Rosenblatt's perceptron is based on the **McCulloch and Pitts neuron** model.

The step function compares this weighted sum to a threshold. If the input is larger than the threshold value, the output is 1; otherwise, it is 0. This is the most common activation function used in Perceptrons are represented by the Heaviside step function:

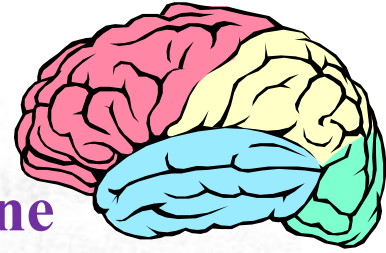
$$h(z) = \begin{cases} 0 & \text{if } z < \text{Threshold} \\ 1 & \text{if } z \geq \text{Threshold} \end{cases}$$

- **Problem:** It cannot learn non-linearly separable functions, such as XOR. If a straight line cannot separate the data points, the perceptron will fail to converge to a solution.

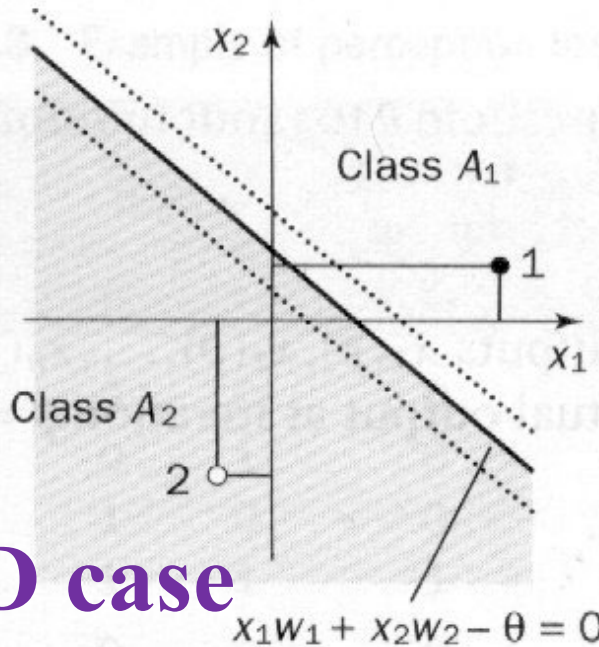
The (single-layer) Perceptron as a linear classifier only



What is the maths ?



hyperline



2-D case

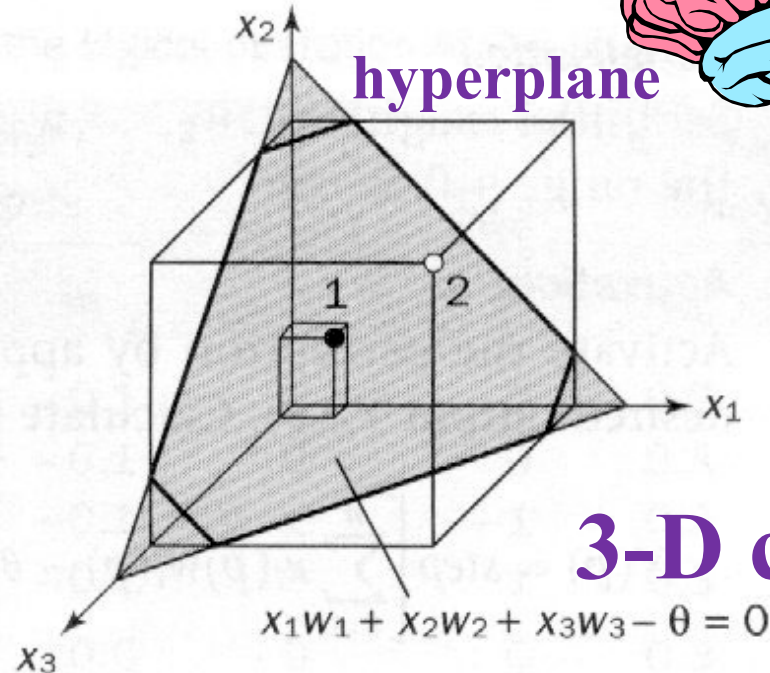
(a)

$$w^t x + b = 0$$
$$w_1 x_1 + w_2 x_2 + b = 0$$
$$w_2 x_2 = -w_1 x_1 - b$$
$$x_2 = \frac{-w_1}{w_2} x_1 - \frac{b}{w_2}$$

.... remember linear equation...

$$y = a x + c$$

hyperplane



3-D case

Conclusion: Perceptron is an algorithm for learning of binary classifiers. It is a type of linear classifier.

- Hence, the n-dimensional space is divided by a hyperplane into **two decision regions**.
- The hyperplane is defined by the linearly separable function

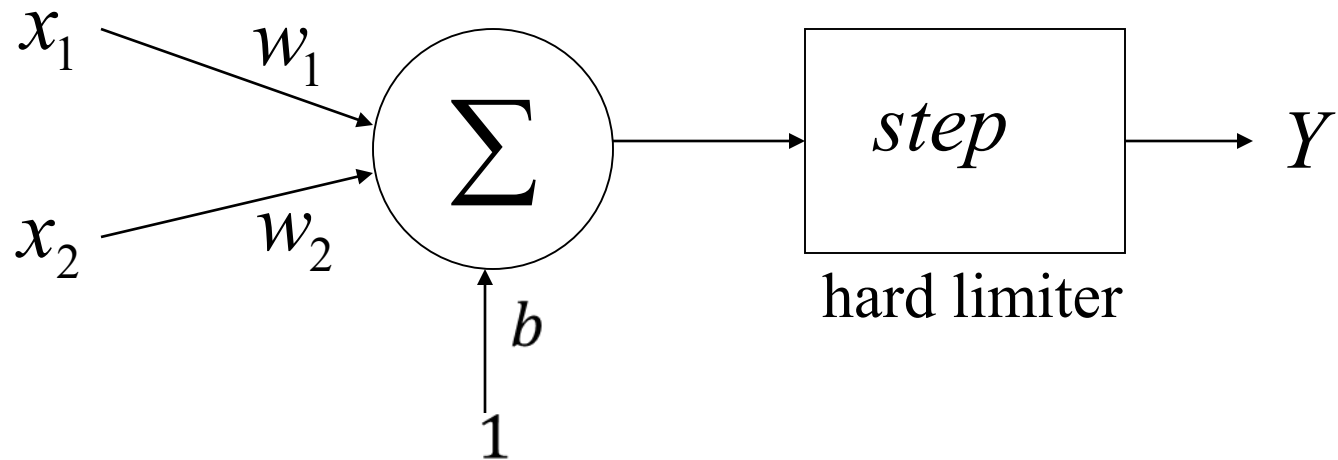
$$\sum_{i=1}^n x_i w_i + b' = 0$$

- For the case of **two inputs**, x_1 and x_2 , the decision boundary takes the form of a **straight line**. The threshold θ can be used to shift the decision boundary.
- For the case of **three inputs** (figure (b)), the **separating plane** is defined by the equation

$$x_1 w_1 + x_2 w_2 + x_3 w_3 + b' = 0$$

A Model of the Perceptron

The equation:
$$Y = \text{step} \left[\sum_{i=1}^n x_i w_i + b \right]$$



The threshold is modeled as an additional input of 1 with weight equals to b .

The Perceptron Learning Rule

for training a perceptron to learn its classification task

- Step 1: Initialization

Set initial weights $w_1, w_2, \dots, w_n$ and threshold θ to random numbers in the range $[-0.5, 0.5]$.

- Step 2: Activation

Activate the perceptron by applying **inputs** $x_1(p), x_2(p), \dots, x_n(p)$ and **desired output** $Y_d(p)$. Calculate the actual output at iteration $p = 1$

$$Y(p) = \text{step} \left[\sum_{i=1}^n x_i(p) w_i(p) + b \right]$$

where n is the number of the perceptron inputs.

- Step 3 : Weight training

Update the weights of the perceptron by

$$w_i(p+1) = w_i(p) + \Delta w_i(p)$$

where $\Delta w_i(p)$ is the weight correction at iteration p.

The weight correction is computed by the **delta**

rule : $\Delta w_i(p) = \alpha \cdot x_i(p) \cdot e(p)$

where $e(p) = Y_d(p) - Y(p)$

Also,

Desired Actual
Output Output from NN

$$b_i(p+1) = b_i(p) + \Delta b_i(p) \quad \text{with} \quad \Delta b_i(p) = \alpha e(p)$$

- Step 4 : Iteration

Increase iteration p by one, go back to Step 2 and repeat the process until convergence.

Can we train a perceptron to perform basic logical operations such as AND, OR or Exclusive-OR ?

Inputs **Desired Output**

Table 6.2 Truth tables for the basic logical operations

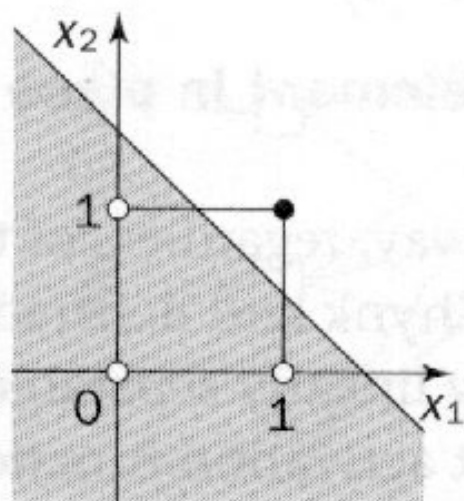
Input variables		AND	OR	Exclusive-OR
x_1	x_2	$x_1 \cap x_2$	$x_1 \cup x_2$	$x_1 \oplus x_2$
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

The perceptron can learn the AND and OR operation.

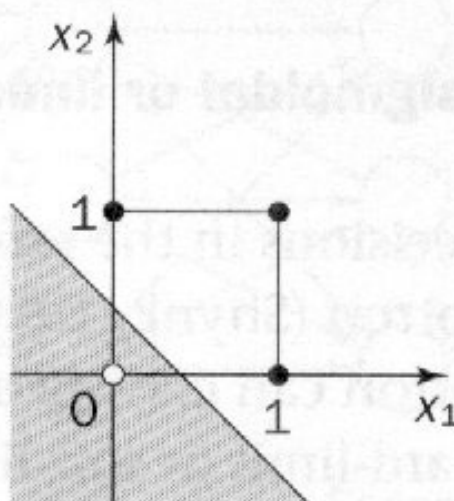
- However, *a perceptron cannot be trained to perform the operation Exclusive-OR.*

A Problem !

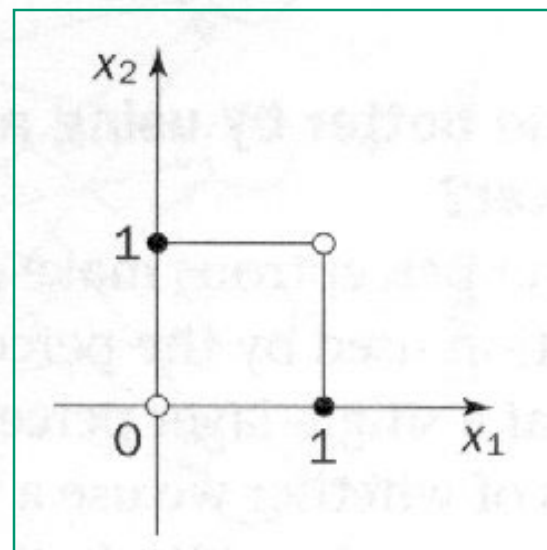
- A little geometry can help us to understand why this is.



(a) AND ($x_1 \cap x_2$)



(b) OR ($x_1 \cup x_2$)



(c) Exclusive-OR
($x_1 \oplus x_2$)

The perceptron can learn the AND and OR operation
But not the XOR.

- The dots in figure (c) are not separable by a single line.
- **A perceptron is able to represent a function only if there is some line that separates all the black dots from all the white dots.** Such functions are called **linearly separable**.
- The fact that a perceptron can learn only linear separable functions is rather bad news. This immense weakness caused the neural network field to stagnate for many years.
- To cope with such problems, we need **multilayer** neural networks.

Limitations of Perceptron

The Perceptron was a significant breakthrough in the development of neural networks, proving that simple networks could learn to classify patterns. However, the Perceptron model has certain limitations that can make it unsuitable for some tasks:

- **Limited to linearly separable problems**
- **Struggles with convergence when handling non-separable data**
- **Requires labeled data for training**
- **Sensitive to input scaling**
- **Lacks hidden layers for complex decision-making**

To overcome these limitations, more advanced neural network architectures, such as Multilayer Perceptrons (MLPs) and Artificial Neural Networks (ANNs), have been developed. These models can learn more complex patterns and are widely used in modern machine learning and deep learning applications.